Submission Summary

Conference Name	11th National Conference on Computer Vision, Pattern Recognition, Image Processing, and Graphics
Paper ID	240
Paper Title	IndoLepAtlas: A Fine-Grained Dataset of Indian Lepidoptera with Geotemporal Metadata and Baseline Classification Study
Abstract	India is home to an extensive range of insect species but is severely underrepresented in global biodiversity datasets. This creates a significant lack of quality training assets for creating region-specific classification architectures. For species of order Lepidoptera (Butterflies & Moths), with visually striking features that change with geography and climate, the problem is even more pronounced. To mitigate this, we created the IndoLepAtlas, a curated 60,641-image, 967-species India-specific butterfly dataset with geotemporal metadata, early life-stage documentation (9,804 images), and 703 images of larval host plant species. The dataset is curated from images submitted to a community-driven source, and is verified over multiple stages, ensuring quality and diversity in butterfly species. Our dataset was also cross-referenced against iNaturalist, a globally renowned biodiversity dataset, revealing a significant butterfly species coverage gap, which IndoLepAtlas intends to bridge. Fine-grained classification of butterfly species, often with very subtle inter-species differences or even great intra-species differences, poses a compounding problem. Our research leveraged a unique deep-learning model with Coordinate Attention (CA), Multi-Level Feature Interaction (MLFI) and geotemporal feature integration. We also demonstrate the effectiveness of our model in accurately classifying butterfly species, showcasing improvements over other commonly used architectures.
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Submission Files	indolepatlas.pdf (9.2 Mb, 2026-04-30 19:17:05 GMT+5:30)